Supplementary Figures and Table for Automated Metabolite Formula Ranking Using Formula Subset Analysis for Characteristic Fragments in Liquid Chromatography–Tandem Mass Spectrometry

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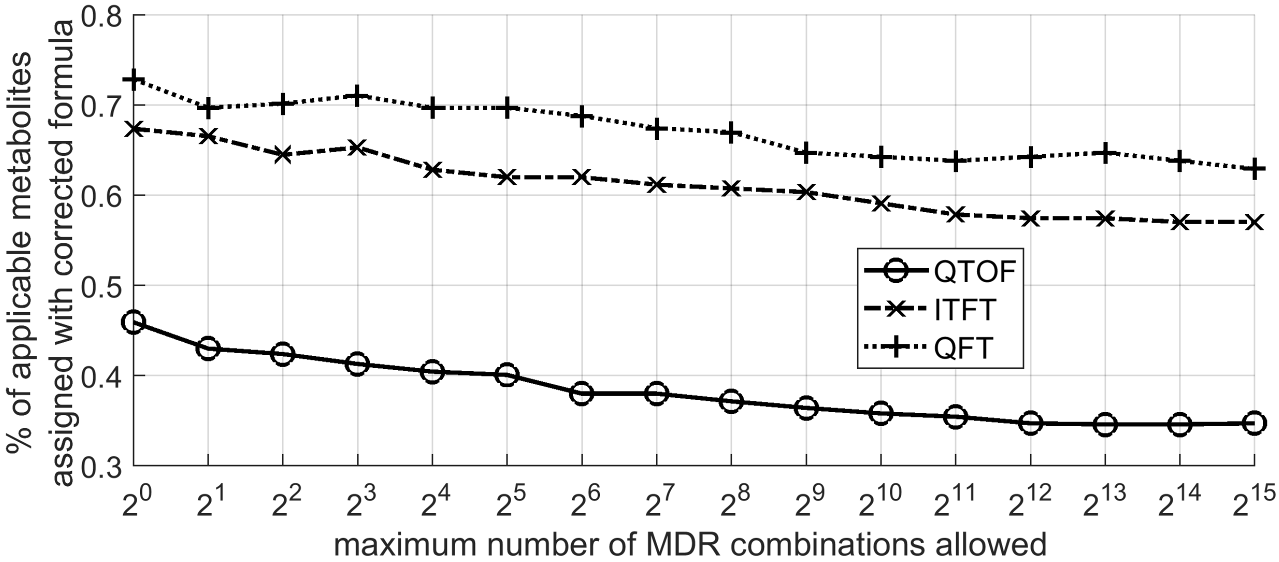


Figure S1. Percentage of applicable metabolites assigned with the correct precursor formula under various limits of MDR combinations (MaxComb) when the mass matching tolerance is 10ppm. The spectra in the HMDB dataset was divided into three different instrument types (QTOF, ITFT, and QFT). The results show that most metabolites were assigned with correct formula when the MaxComb was set to one, which is consistent in spectra acquired from different instruments.

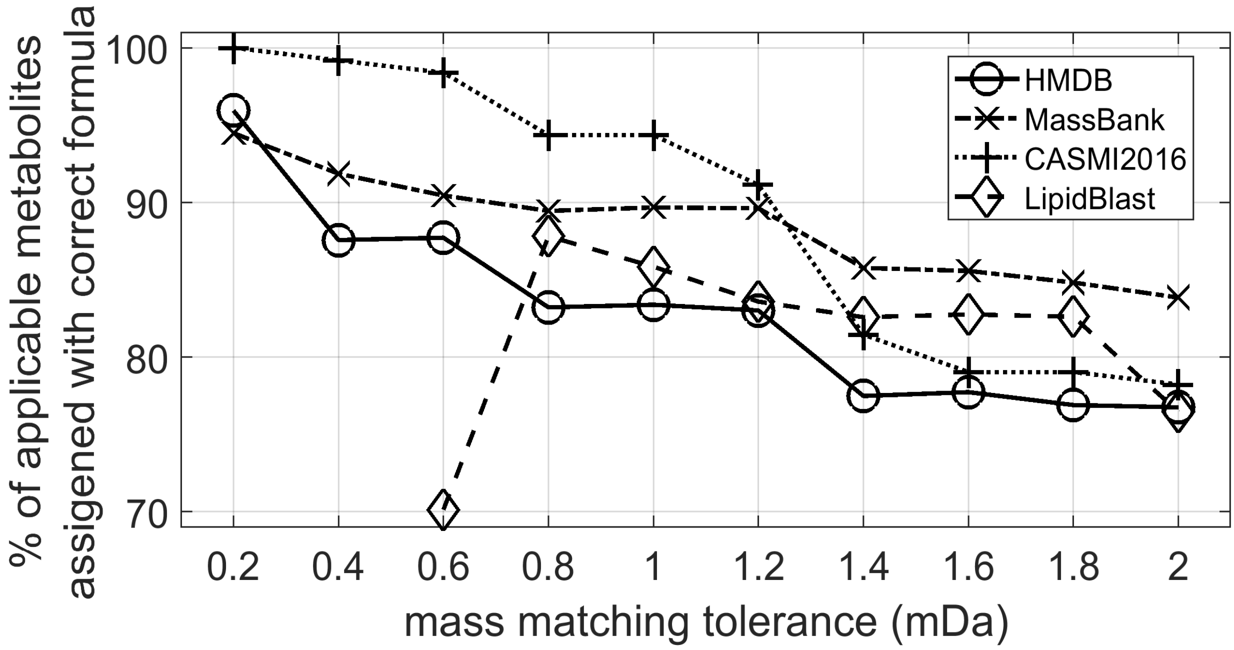


Figure S2. Percentage of applicable metabolites assigned with the correct formula in the four datasets at the mass matching tolerance range 0.2–2 mDa. Note that the accuracy of the precursor mass data in LipidBlast is not higher than 0.8 mDa and thus no correct formula can be found beyond this tolerance.

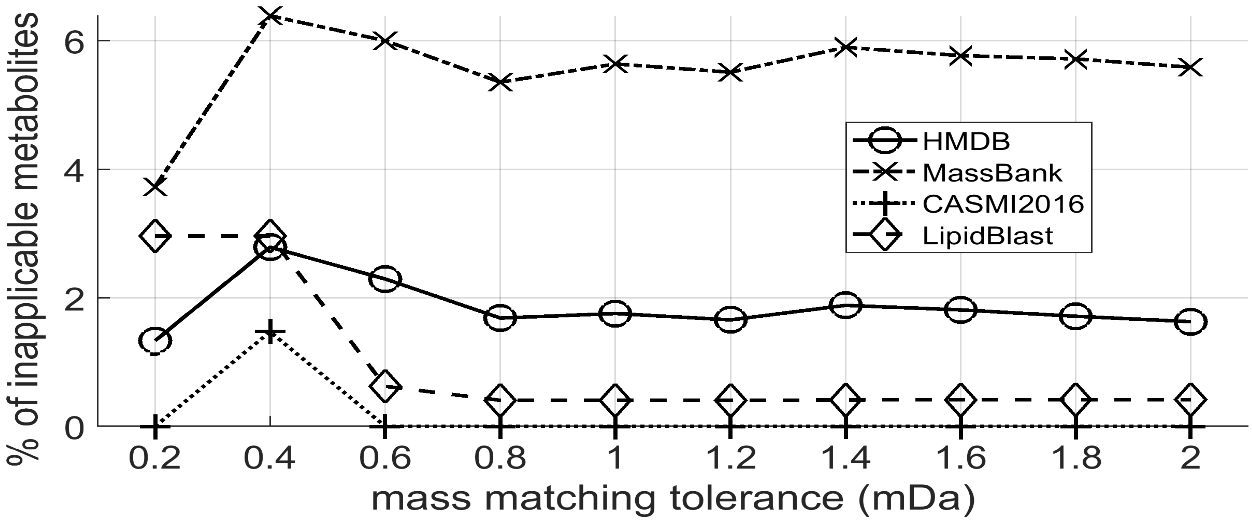


Figure S3. Percentage of inapplicable metabolites in the four datasets in the mass matching tolerance range 0.2–2 mDa. An inapplicable metabolite means that the mass difference between the precursor in its spectrum and its theoretical value is larger than the tolerance, so that the correct metabolite cannot be found. The result shows that the percentage of inapplicable metabolites reached relatively lower values when the tolerance was set to the range of [0.8, 1.2].

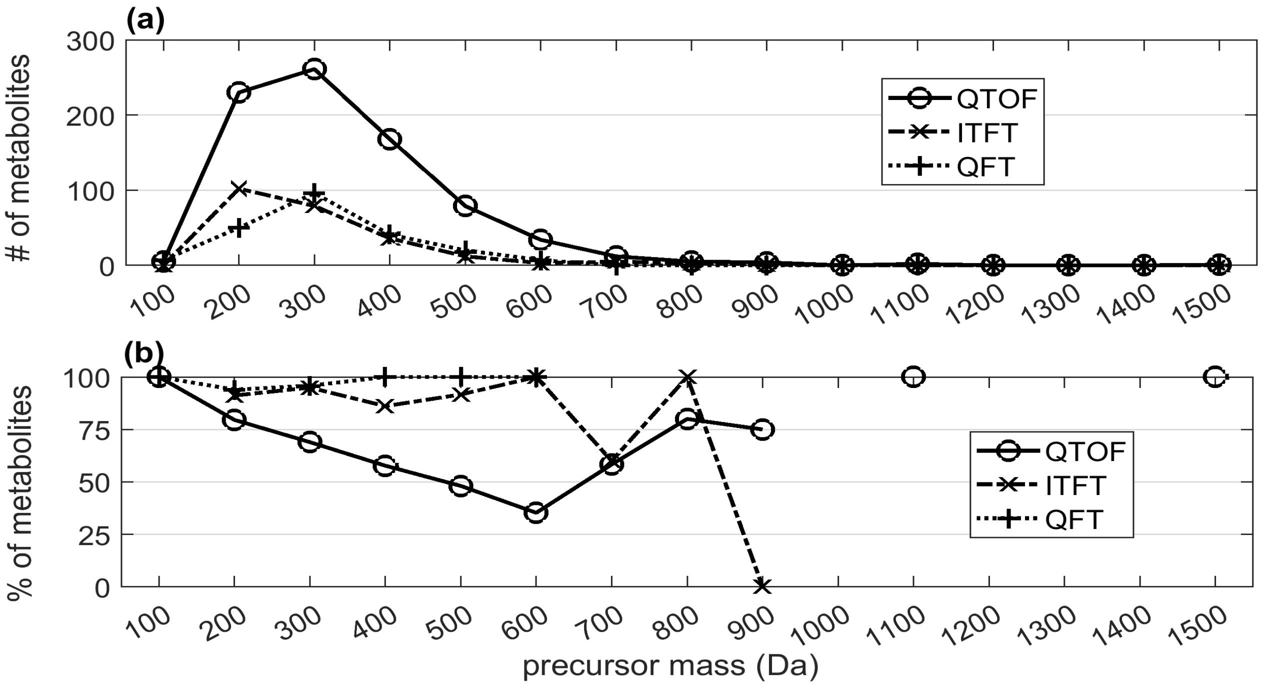


Figure S4. Performance of FSA for MS/MS spectra at the various precursor mass intervals in the HMDB dataset. The performance is presented by the (a) number and (b) percentage of metabolites whose correct precursor formula was found rank#1 by FSA. The spectra were divided into three groups (QTOF, ITFT, and QFT) based on the instrument type where they were acquired. The mass accuracy in the three instruments are QFT > ITFT > QTOF. The masses of the largest metabolites are 1442.33, 836.52, and 563.30 Da in the QTOF, ITFT and QFT dataset, respectively. In addition, the missing points are attributed to the absence of a metabolite in the corresponding mass intervals.

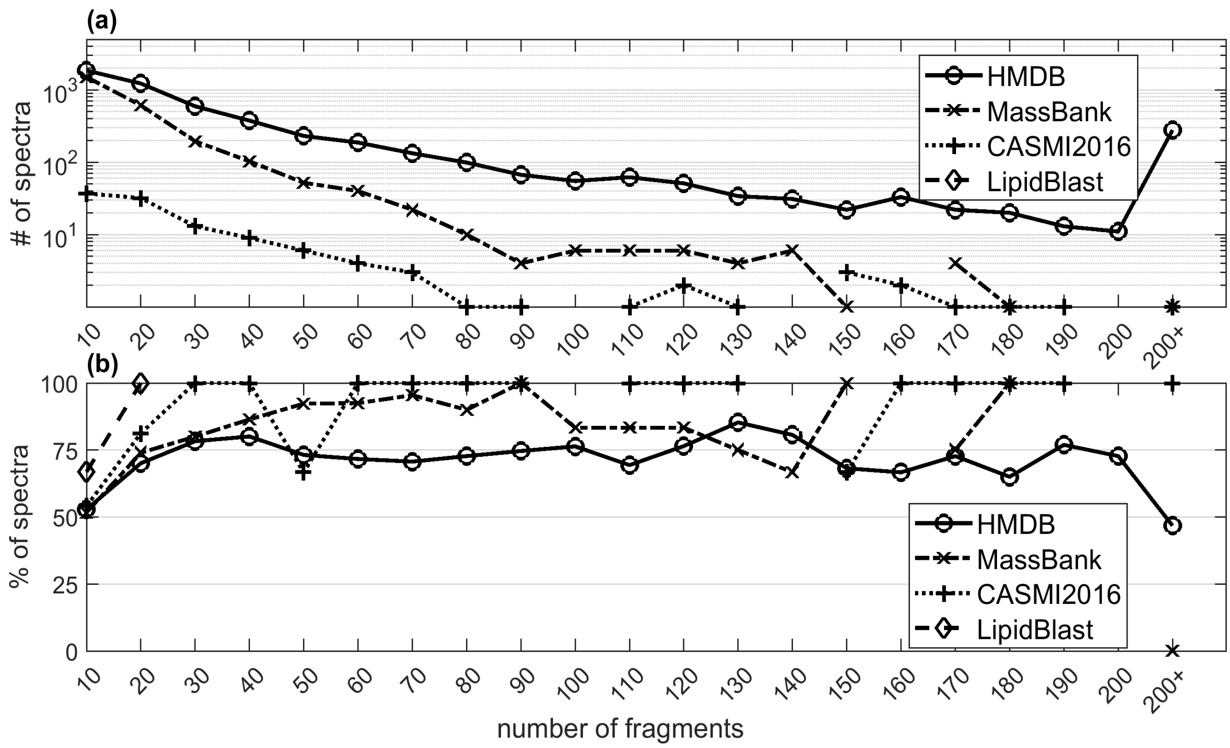


Figure S5. Performance of the FSA method at various fragment numbers in the four datasets. The performance is represented by the (a) number and (b) percentage of spectra whose correct precursor formula was found rank#1 by FSA. The maximum numbers of fragments in the applicable spectra are 1502, 267, 218, and 16 in the HMDB, MassBank, CASMI2016, and LipidBlast datasets, respectively. Because the number of spectra with more than 200 fragments is rather small, these spectra are grouped together under 200+ fragments.

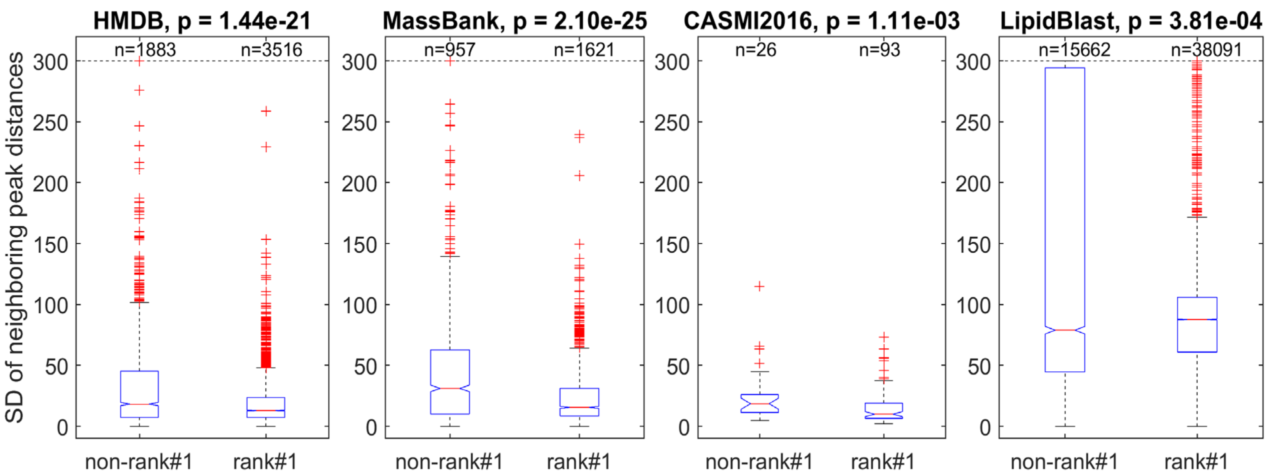


Figure S6. Boxplots of the fragment distance standard deviations (SDs) in the rank#1 spectra (spectra which formula was solely ranked first) versus those in the other spectra for the three datasets. The Wilcoxon rank sum test showed that for all three datasets, the rank#1 spectra exhibited significantly ($p < 0.01$) smaller SDs in neighboring peak distances than did the other spectra. Note: A few spectra with SDs >300 were omitted in the QTOF and ITFT data sets for a clear view of the corresponding boxplots.

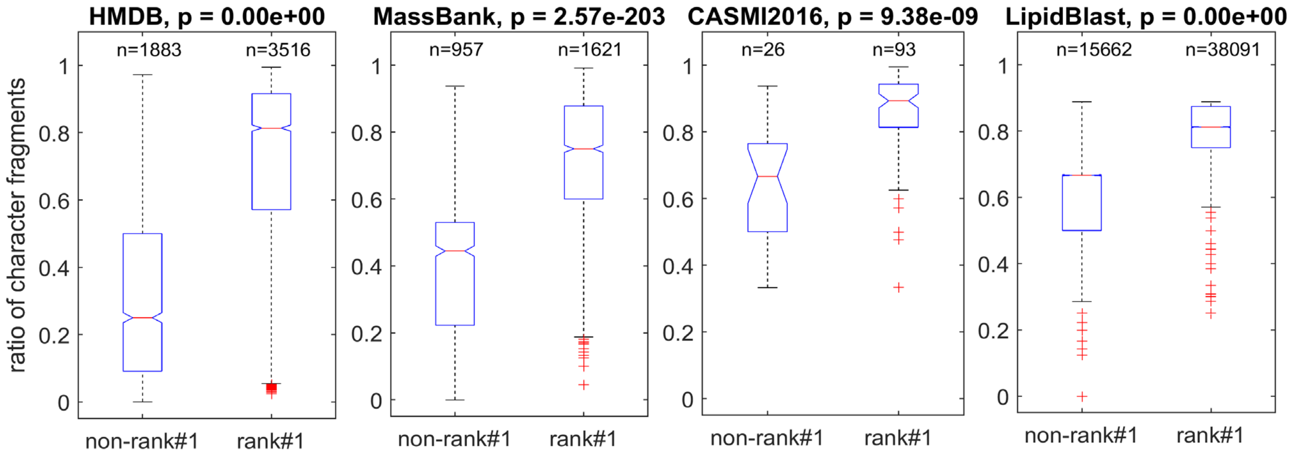


Figure S7. Boxplot of ratios of the CFs in the rank#1 spectra (spectra which formulas solely ranked first) versus those in the non-rank#1 spectra for the three datasets. The Wilcoxon rank sum test indicates that the rank#1 spectra exhibit significantly ($p < 0.01$) higher CF ratios than those in the other spectra in all three datasets.

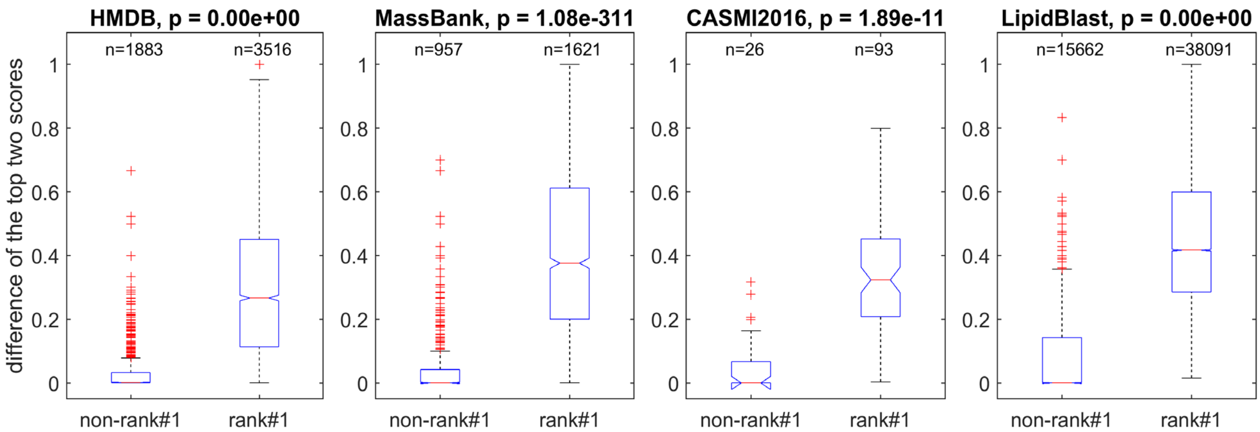
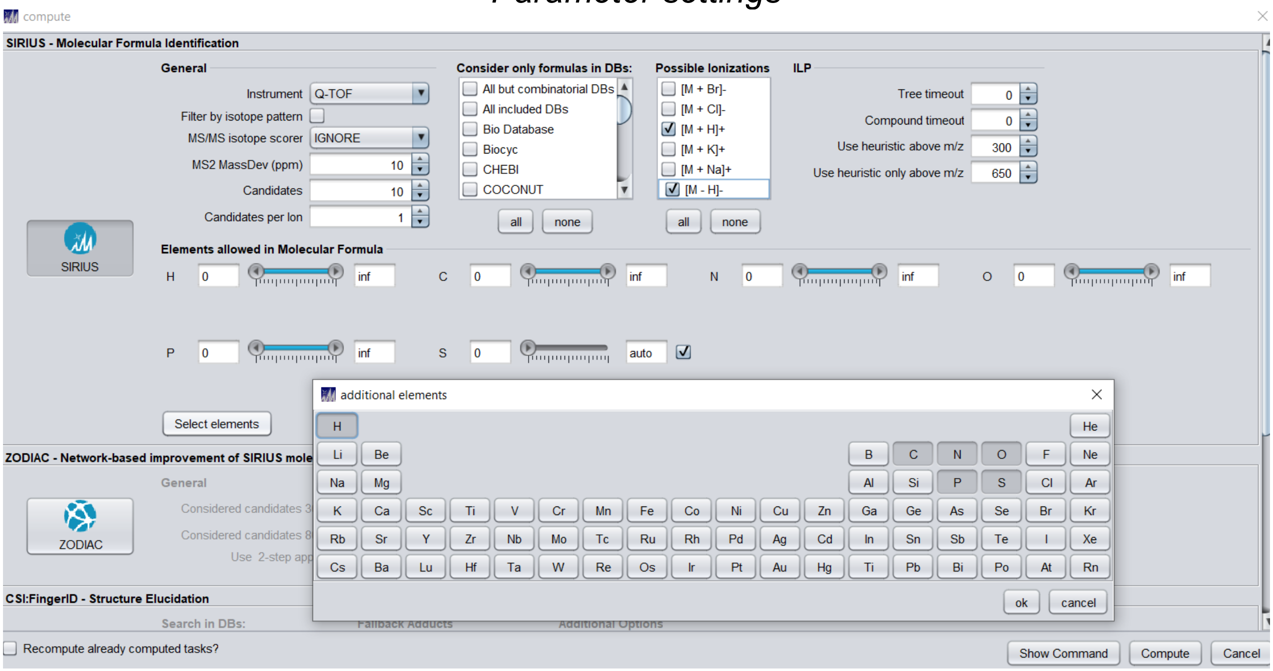


Figure S8. Boxplot of the top two score differences in the rank#1 spectra (spectra which formulas solely ranked first versus those in the non-rank#1 spectra for the three datasets. The Wilcoxon rank sum test indicates that the rank#1 and non-rank#1 spectra exhibit significant ($p < 0.01$) score differences in all three datasets.

Table S1. Test result of SIRIUS 4.0 on the HMDB dataset

Dataset	*Precision*	*Number of spectra**	*Number of errors*	*Number of spectra Not processed*	*Time cost (Records per sec)*
QTOF	0.5776	2770	1157	13	3.7432
QTF	0.6912	1402	433	0	5.3923
ITFT	0.6718	2328	764	0	2.8135

Parameter settings



Specifications of the tested computer

CPU: Intel(R) Core(TM) i7-7700 CPU @ 3.60GHz   3.60 GHz RAM: 32.0 GB

* The input spectra with the same m/z values of precursor and fragments were merged.